

2025-2026 – Econ 0039 – Advanced Macro

Lecture 3 : Income and Wealth Inequalities

Franck Portier

f.portier@ucl.ac.uk

University College London

Version 1.1

02/02/2026

Changes from version 1.0

Disclaimer

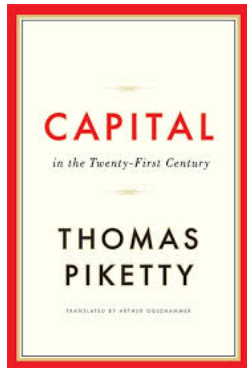
These are the slides I am using in class. They are not self-contained. They do contain original material and some that is not, including some “cut and paste” from various sources that I am citing as much as possible. These slides are not intended to be used outside of the course nor to be distributed. Thank you for signalling me typos or mistakes at f.portier@ucl.ac.uk.

0. Introduction

What is this lecture about ?

- ▶ In 2014, extraordinary success of THOMAS PIKETTY's book *Capital in the Twenty-First Century*
- ▶ Main questions
 - × How did inequalities evolved over the last century ? (income, wealth, capital income)
 - × What can we expect for the future ?
 - × $r > g$?
- ▶ According to RICARDO in the preface of his *Principles of Political Economy and Taxation* (1817), the determination of the laws of distribution is :
“the principal problem in Political Economy.”

Figure 1 – PIKETTY'S BOOK COVER



0. Introduction

Plan of the Lecture

1. Income distribution in history of economic thought
2. KUZNETS' Facts
3. PIKETTY's Facts
4. PIKETTY's two "laws of capitalism"
5. $\frac{k}{y}$ in the SOLOW model and beyond
6. The elasticity of substitution between k and ℓ
7. Summary
8. References

1. Income distribution in history of economic thought

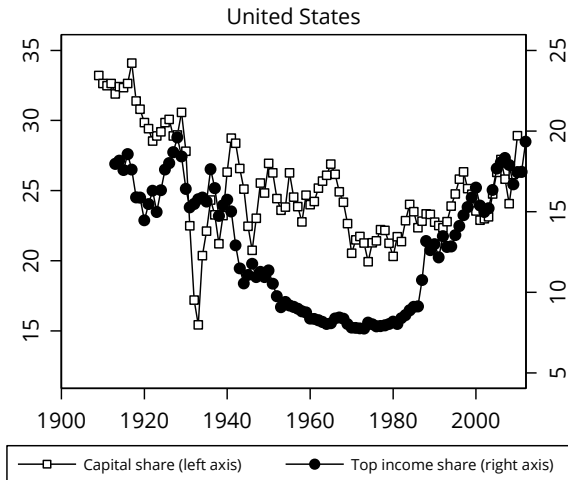
Income distribution

- ▶ Two distinct concepts
 - × *Functional* distribution of income, *i.e.* the distribution of income between the main factors of production : capital, labor, land
 - × *Personal* distribution of income between individuals : wage inequalities, capital or land ownership
- ▶ Two different level of analysis
 - × *Positive* : Why are some people rich and some poor ?
 - × *Normative* : Which level of inequalities is justified ?
- ▶ I am discussing only the **positive** aspect and mainly **functional** distribution.
- ▶ Note that capital income is quite unevenly distributed, so that there is a link between functional and personal inequalities.

1. Income distribution in history of economic thought

From functional to personal inequalities

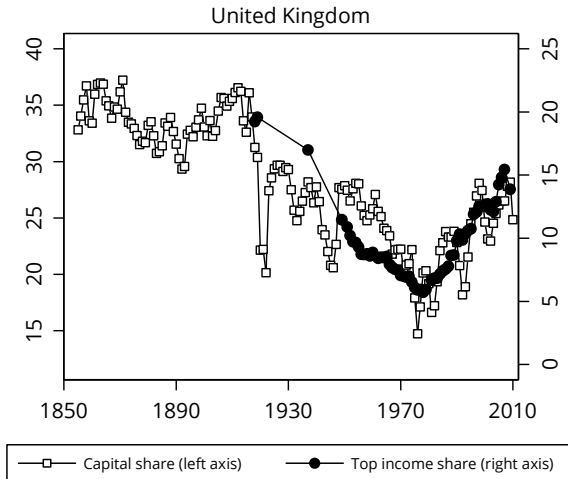
Figure 2 – CAPITAL SHARE IN TOTAL INCOME AND TOP 1% INCOME SHARE (BENGTTSSON AND WALDENSTRÖM (2015))



1. Income distribution in history of economic thought

From functional to personal inequalities

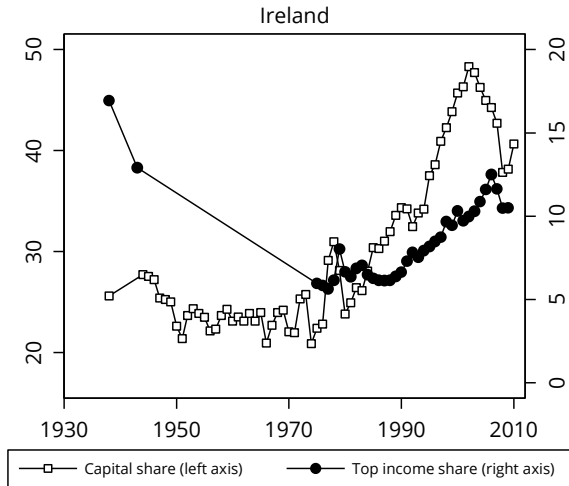
Figure 3 – CAPITAL SHARE IN TOTAL INCOME AND TOP 1% INCOME SHARE (BENGTTSSON AND WALDENSTRÖM (2015))



1. Income distribution in history of economic thought

From functional to personal inequalities

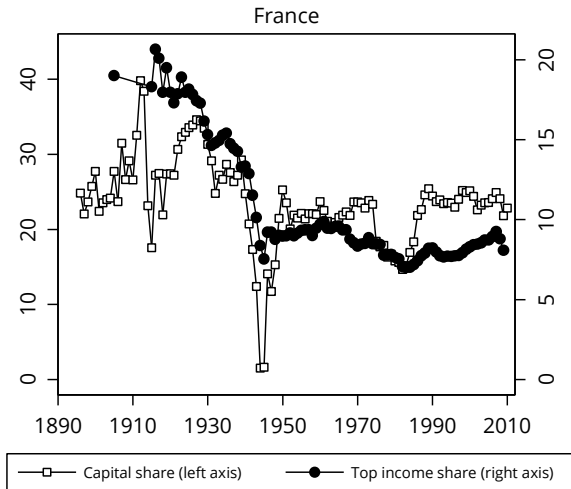
Figure 4 – CAPITAL SHARE IN TOTAL INCOME AND TOP 1% INCOME SHARE (BENGTTSSON AND WALDENSTRÖM (2015))



1. Income distribution in history of economic thought

From functional to personal inequalities

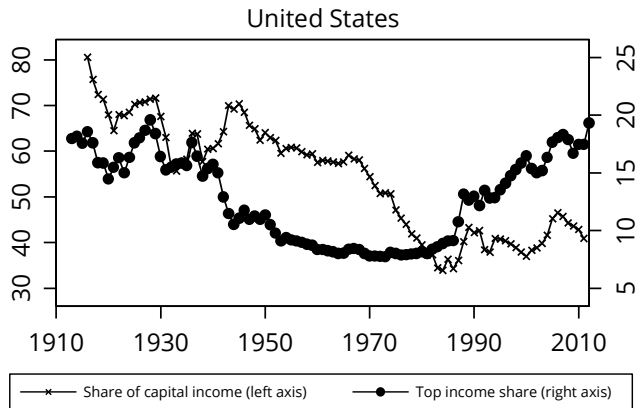
Figure 5 – CAPITAL SHARE IN TOTAL INCOME AND TOP 1% INCOME SHARE (BENGTTSSON AND WALDENSTRÖM (2015))



1. Income distribution in history of economic thought

From functional to personal inequalities

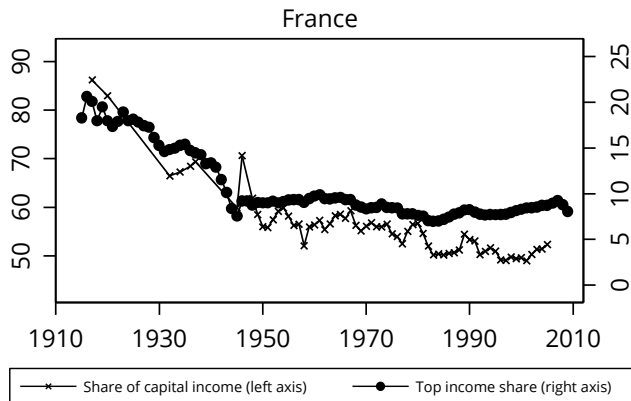
Figure 6 – CAPITAL SHARE IN TOP 1% INCOME AND TOP 1% INCOME SHARE
(BENGTTSSON AND WALDENSTRÖM (2015))



1. Income distribution in history of economic thought

From functional to personal inequalities

Figure 7 – CAPITAL SHARE IN TOP 1% INCOME AND TOP 1% INCOME SHARE
(BENGTTSSON AND WALDENSTRÖM (2015))

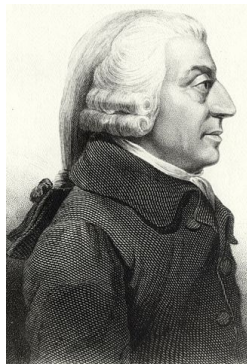


1. Income distribution in history of economic thought

ADAM SMITH (1723-1790)

- ▶ 1776 : *An Inquiry into the Nature and Causes of the Wealth of Nations*
- ▶ Division of labor makes it possible to increase labor productivity
 - × Pin factory
 - × The production of a pin has been broken down into “about eighteen” separate operations,
 - × Productivity is multiplied by 4800.
- ▶ but no proportional increase in the wage... why?
- ▶ Main reason : any increase in the general level of wages would lead to an increase of population and therefore of the work force, and this would tend to reverse the initial increase of wages.
- ▶ Typical effect in the “malthusian regime”

Figure 8 – ADAM SMITH



1. Income distribution in history of economic thought

DAVID RICARDO (1772-1823)

- ▶ 1817 : *Principles of Political Economy and Taxation*
- ▶ Wages are explained by the Malthusian dynamics already described by ADAM SMITH
- ▶ Profits (capital income) is the rate of return on capital, defined as the rate of interest plus a risk premium.
- ▶ The other really important factor (besides labor) is land.
- ▶ Rent is the income that goes to land owners.

Figure 9 – DAVID RICARDO



1. Income distribution in history of economic thought

DAVID RICARDO on rent

- ▶ *Scarcity principle*
- ▶ Land varies in terms of its quality or productivity.
- ▶ RICARDO's generic term for agricultural products : *corn*
- ▶ The price of corn is determined by the cost of labour and capital required to produce a unit of corn on the land with the lowest quality.
- ▶ Corn has same quality on all types of land $\rightsquigarrow$ all corn will sell at the same price
- ▶ Therefore, a positive rent will exist on all inframarginal units of land.

1. Income distribution in history of economic thought

DAVID RICARDO on rent (*continued*)

- ▶ Rent is determined by the cost of labour and capital used on the margin of cultivation,
- ▶ and the position of this margin is determined by the price of corn.
- ▶ *“Corn [price] is not high because a rent is paid, but a rent is paid because corn [price] is high”, RICARDO (1817)*
- ▶ What is the dynamics of a growing economy (for example with technological progress) ?
 - × wages increase above the level of subsistence $\rightsquigarrow$ population expands $\rightsquigarrow$ demand for corn increases $\rightsquigarrow$ the margin of cultivation is extended.
 - × The share of rent and wages in national income go up
 - × even after the wage rate has returned to its level of subsistence (because population is larger).
 - × The implication of this is that profits fall
 - × weakening of the incentive to invest $\rightsquigarrow$ the economy reaches its stationary state,

1. Income distribution in history of economic thought

DAVID RICARDO's principle of scarcity

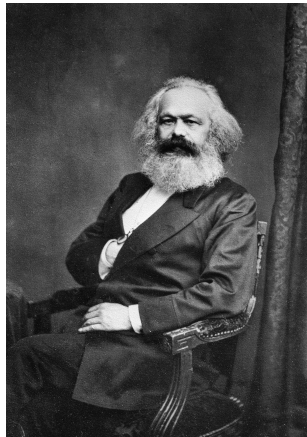
- ▶ Land tends to become increasingly scarce relative to other goods.
- ▶ Land owners will therefore claim a growing share of national income, as the share available to the rest of the population decreases
- ▶ Risks of social and political instability.
- ▶ For Ricardo, the only logically and politically acceptable answer was to impose a steadily increasing tax on land rents.
- ▶ The Ricardian principle of scarcity is still important to understand the distribution of wealth in the twenty-first century.
- ▶ Replace the price of farmland in RICARDO's model by
 - × the price of urban real estate in major world capitals
 - × the price of oil.

1. Income distribution in history of economic thought

KARL MARX (1818-1883)

- ▶ “Das Kapital” (1867-1894)
- ▶ MARX positive theory of distribution is broadly consistent with SMITH and RICARDO.
- ▶ MARX adopts the theory of subsistence wages
- ▶ and adds unemployment (“*industrial reserve army*”)
- ▶ The existence of this reserve army is in fact in the interest of the capitalists
- ▶ Because there are fluctuations in economic activity that also imply large fluctuations in the demand for labor.
- ▶ A distinctive feature of MARX is the concept of *exploitation*

Figure 10 – KARL MARX



1. Income distribution in history of economic thought

KARL MARX's concept of exploitation

- ▶ Labour is the fundamental factor of production in the sense that all non-labour inputs can be derived from past labour.
- ▶ *“all commodities are only definite masses of congealed labour time”*, “Das Kapital”, 1867-1894
- ▶ But workers are only paid the subsistence wage
- ▶ The difference between the two is the worker's unpaid work for the benefit of the capitalist.
- ▶ This is the profit or surplus value which defines the capitalist's exploitation of the worker.

1. Income distribution in history of economic thought

Dynamics of capitalist economies for MARX

- ▶ Capital is primarily industrial (machinery, plants, etc.) rather than land property,
- ▶ In principle there is no limit to the amount of capital that could be accumulated.
- ▶ $\rightsquigarrow$ inexorable tendency for capital to accumulate and become concentrated in ever fewer hands.
- ▶ two alternative ends :
 - × the rate of return on capital steadily diminishes $\rightsquigarrow$ kills the engine of accumulation and leads to violent conflict among capitalists,
 - × capital's share of national income increases indefinitely $\rightsquigarrow$ social revolts

1. Income distribution in history of economic thought

The marginalist approach

- ▶ Marginalists seek to anchor their analysis in the behaviour of individual economic agents, using the theory of optimization and the mathematical tools of the differential calculus.
 - × WILLIAM STANLEY JEVONS : "Theory of Political Economy" (1871)
 - × CARL Menger : "Grundsätze der Volkswirtschaftslehre" (1871)
 - × LÉON WALRAS : "Éléments d'économie politique pure" (1874-1877).
- ▶ Essentially brings back demand (coming from marginal utility) in the theory of value (and not only the cost of production)
- ▶ Returns of labor, land and capital are derived from supply and demand
- ▶ WALRAS criticized Malthus for the lack of logic in his theory of population, because technological progress is neglected...
- ▶ ...and points out the failure of Malthusian theory to explain the actual increase in living standards for all classes in society.

Figure 11 –
JEVONS, Menger
& WALRAS



1. Income distribution in history of economic thought

Modern Neoclassical Theory

- ▶ Consider the SOLOW model as illustrative of the modern theory.
- ▶ We will review this model later.
- ▶ Constant saving rate + Neoclassical production function
- ▶ Exogenous technical progress
- ▶ Existence of a Balance Growth Path along which the shares of capital and labor are constant.
- ▶ Sustained growth without (functional) inequalities explosion.

1. Income distribution in history of economic thought

Dynamics

- ▶ Classics and MARX : Stationary state with subsistence wage or collapse
- ▶ Neoclassical economics : Balanced growth is possible (SOLOW)
- ▶ All this theory has been developed with virtually no data ... until KUZNETS in the 1950's.

Figure 12 – SIMON KUZNETS



Simon Kuznets

2. KUZNETS' Facts

Method

- ▶ SIMON KUZNETS : American economist (Russian immigrant, 1901-1985)
- ▶ 1953 book : *"Shares of Upper Income Groups in Income and Savings"*
- ▶ United States from 1913 to 1948
- ▶ First publication that uses two sources of data
 - × US federal income tax returns (which did not exist before the creation of the income tax in 1913)
 - × US national accounts
- ▶ KUZNETS can compute the evolution of the share of each decile, as well as of the upper centiles, of the income hierarchy in total US national income.

2. KUZNETS' Facts

Findings

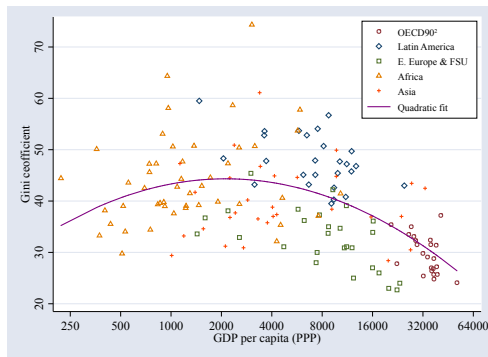
- ▶ Sharp reduction in income inequality in the United States between 1913 and 1948
- ▶ The upper decile of the income distribution went down from 45-50 % of annual national income to 30-35 %
- ▶ This 10% of income reduction is equal to half of the income of the poorest 50 percent of Americans
- ▶ Bottom line : Inequalities are reducing

2. KUZNETS' Facts

KUZNETS curve

- ▶ 1955 *"Economic Growth and Inequality"* (Econometrica)
- ▶ Inequality everywhere can be expected to follow a bell curve.
- ▶ It should first increase and then decrease over the course of industrialization and economic development :
 - × first phase of naturally increasing inequality associated with the early stages of industrialization (US in the XIXth century)
 - × second phase of sharply decreasing inequality (US in the beginning of the XXth century)

Figure 13 – A KUZNETS CURVE IN 1990



"Is there a Kuznets curve?", J. Gallup [2012]

3. PIKETTY's facts

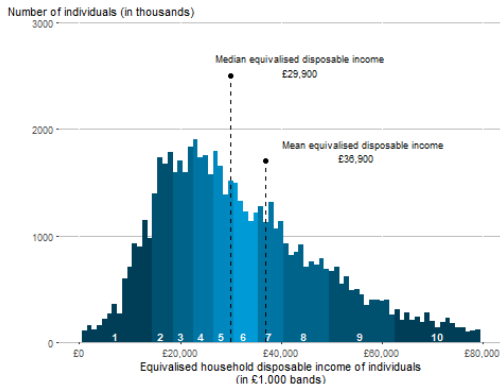
Capital in the Twenty-First Century

- ▶ THOMAS PIKETTY book published in 2014
- ▶ He puts together data on income and wealth inequalities for many countries over a long period
- ▶ Here I consider U.S. and Europe
- ▶ Piketty shows that that the KUZNETS observation cannot be extrapolated and that there are no obvious KUZNETS curves.
- ▶ Three key observations related to
 - × Income inequalities
 - × Wealth inequalities
 - × Wealth-to-Income Ratios
- ▶ I will use results from PIKETTY AND SAEZ (2014) in *Science*.
- ▶ Restrict to US and Europe

3. PIKETTY's facts

Remark : A distribution cannot be summarised by one number

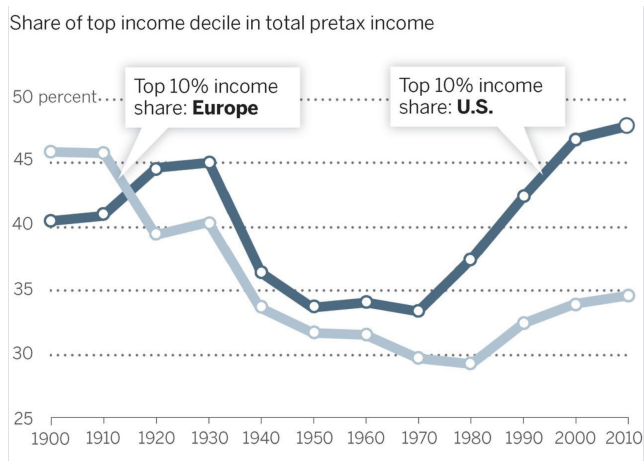
Figure 14 – DISTRIBUTION OF UK HOUSEHOLD DISPOSABLE INCOME, FINANCIAL YEAR ENDING 2020 (ONS,HOUSEHOLD FINANCES SURVEY)



3. PIKETTY's facts

Income inequalities

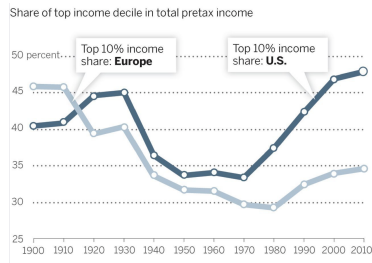
Figure 15 – INCOME INEQUALITY IN EUROPE AND THE UNITED STATES, 1900-2010
(PIKETTY AND SAEZ (2014))



3. PIKETTY's facts

Income inequalities

- ▶ Whereas income inequality was larger in Europe than in the United States a century ago, it is currently much larger in the United States.
- ▶ The top decile share in Europe is currently almost one-third smaller than what it used to be one century ago.
- ▶ The secular decline in inequality would be even larger if one takes into account the rise of taxes and transfers, and measure instead income after taxes and transfers.
- ▶ Primary income concentration was in 2010 higher than it has ever been in U.S. history. It is also slightly higher than in pre-WWI Europe.

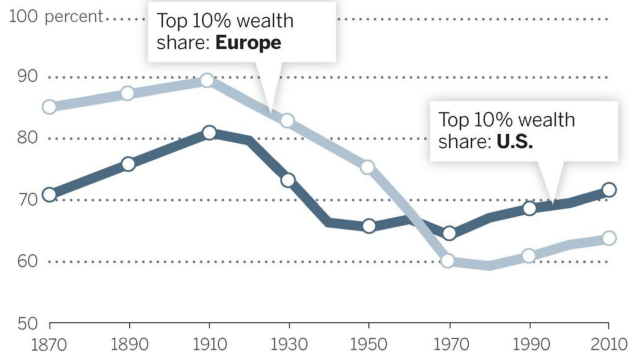


3. PIKETTY's facts

Wealth inequalities

Figure 16 – WEALTH INEQUALITY IN EUROPE AND THE UNITED STATES, 1870-2010
(PIKETTY AND SAEZ (2014))

Share of top wealth decile in total net wealth

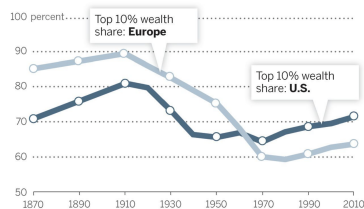


3. PIKETTY's facts

Wealth inequalities

- ▶ Same “great inequality reversal” between Europe and the United States when one looks at wealth inequality rather than income inequality.
- ▶ However, important differences between income and wealth inequality dynamics.
- ▶ Income *versus* wealth :
 - ✗ wealth concentration is always much higher than income concentration.
 - ✗ the bottom 50% wealth share is always less than 5%, whereas the bottom 50% income share generally falls in the 20 to 30% range

Share of top wealth decile in total net wealth

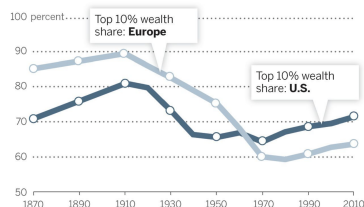


3. PIKETTY's facts

Wealth inequalities – US *versus* Europe

- ▶ In contrast to income inequality, U.S. wealth inequality levels have still not reached the record levels observed in Europe before World War I.
- ▶ Wealth concentration was extreme in pre-WWI Europe : the top decile wealth share was close to 90%, so that the middle 40% wealth holders were almost as poor as the bottom 50% wealth holders (the wealth share of both groups was close to or less than 5%) $\rightsquigarrow$ almost not *middle class*
- ▶ Between 1914 and the 1950s-1960s, the top decile wealth share fell dramatically in Europe, from about 90% to less than 60%. It has been rising since the 1970-80s, and is now close to 65% . $\rightsquigarrow$ the wealth middle class now commands a larger share of total wealth in Europe than in the United States.

Share of top wealth decile in total net wealth

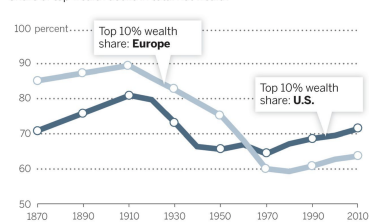


3. PIKETTY's facts

Wealth inequalities – US *versus* Europe

- ▶ Given that wealth inequality is lower in the United States today than in 1913 Europe, why is U.S. income inequality now as large as (or even slightly larger than) that in 1913 Europe?
- ▶ Because modern U.S. inequality is based more on a very large rise of top labor incomes than upon the extreme levels of wealth concentration.

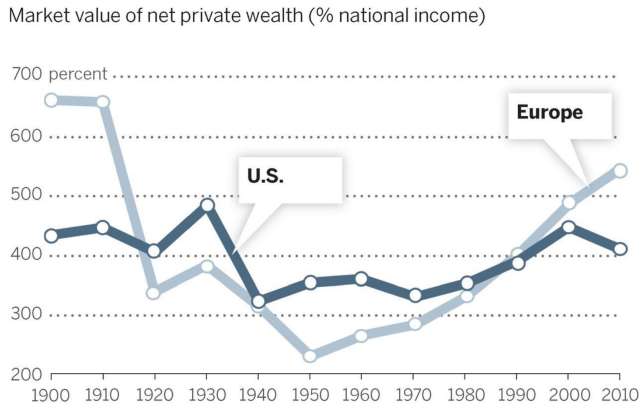
Share of top wealth decile in total net wealth



3. PIKETTY's facts

Wealth-to-Income Ratios

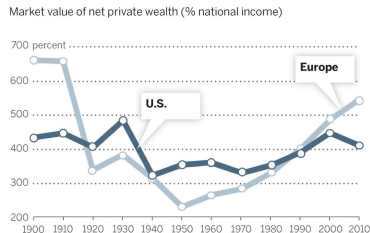
Figure 17 – WEALTH-TO-INCOME RATIOS IN EUROPE AND THE UNITED STATES, 1900-2010 (PIKETTY AND SAEZ (2014))



3. PIKETTY's facts

Wealth-to-Income Ratios

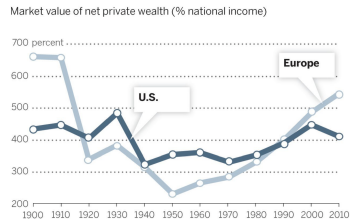
- ▶ Europe aggregate wealth-income ratio : *U-shaped* pattern over the past century :
 - × Eve of WWI, net private wealth was about equal to 6 to 7 years of national income
 - × Fall to about 2 to 3 years of national income in the 1950s.
 - × Regular increase since then, back to about 5 to 6 years of national income.
- ▶ In contrast, the U.S. pattern is flatter, around 4 to 5 years of national income
- ▶ The United States is the land of booming top labor incomes ; Europe is the land of booming wealth.



3. PIKETTY's facts

Wealth-to-Income Ratios – Historical interpretation

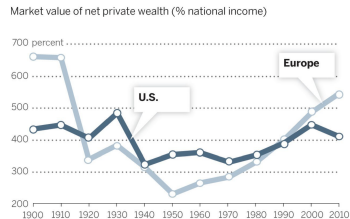
- ▶ Fall of European wealth-income ratios caused by three main factors :
 - ✗ direct war-related physical destruction of domestic capital assets (real estate, factories, machinery, equipment)
 - ✗ lack of investment (a large fraction of 1914-1945 private-saving flows was absorbed by the public deficits induced by war financing)
 - ✗ fall in relative asset prices (real estate and stock market prices were both historically very low in the immediate postwar period, partly due to rent control, nationalization, capital controls, and various forms of financial repression policies).
- ▶ France and Germany : each of these three factors account for about one-third of the total decrease in wealth-income ratios
- ▶ UK : domestic capital destruction is less than 10% of the total); two other factors about half each.



3. PIKETTY's facts

Wealth-to-Income Ratios – Historical interpretation

- ▶ Why was the rise of European wealth-income ratios slow after WWII?
- ▶ Because capital accumulation is a slow process :
- ▶ With a saving rate of 10% per year, it takes 50 years to accumulate the equivalent of 5 years of income.



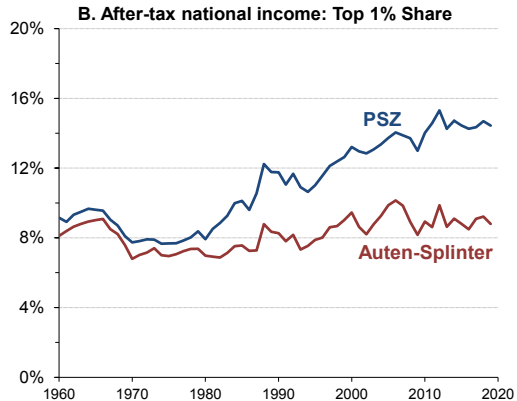
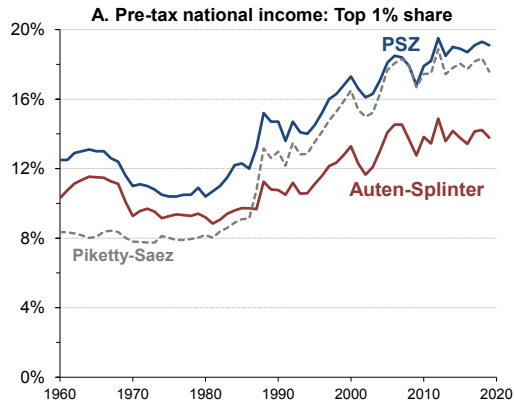
3. PIKETTY's facts

Debates

- ▶ Measuring inequalities is a daunting task.
- ▶ Why isn't it enough to look at individuals tax returns ?
 - × Hard to properly measure investment returns, income from being a partner in a company,...
 - × Some income is not reported : employer-provided benefits and government welfare,...
 - × Also tax evasion varies with time (less use of cash) and income level ("tax optimization" is only for high income)
- ▶ Recent work of AUTEN & SPLINTER "Income Inequality in the United States : Using Tax Data to Measure Long-Term Trends" (Journal of Political Economy [2024]).
 - × Tax reform in 1986 in the US : lower taxes for high income levels + less possibility to use tax shelters $\rightsquigarrow$ more reported income for the rich, although income hasn't changed.
 - × Plus many other adjustment
- ▶ Provocative result : After-tax share of top 1% hasn't increase.
- ▶ Seems to be quite a lower bound $\rightsquigarrow$ debate still going on.

3. PIKETTY's facts

Debates : The findings of AUTEN & SPLINTER



Note : PSZ = PIKETTY, SAEZ & ZUCMAN

4. PIKETTY's two "laws of capitalism"

PIKETTY's view about the long run

- ▶ "First law of capitalism" (actually an identity) :

$$\text{share of capital in total income} = \sigma_k = r \times \frac{k}{y}$$

- ▶ "Second law of capitalism" : if the saving rate s is constant and g is the growth rate of the economy :

$$\frac{k}{y} = \frac{s}{g}$$

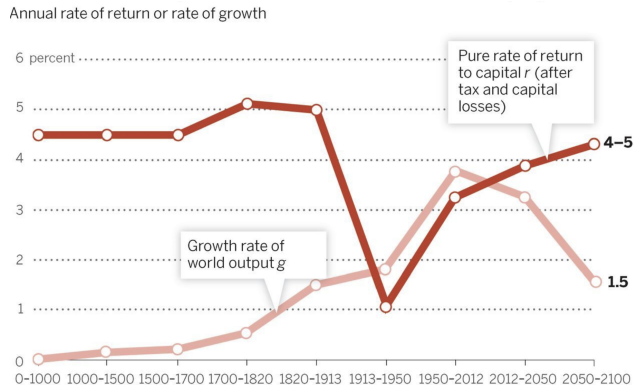
When growth goes to zero, capital becomes arbitrarily bigger than income.

- ▶ Those two laws imply $\sigma_k = s \frac{r}{g}$
- ▶ If g drops while r does not much, then the share of capital in total income becomes very large $\rightsquigarrow$ we shall expect a lot of inequalities ...
- ▶ As indeed $r > g$ (see Figure 18), capital taxes are needed if one dislikes inequalities (now we are on a normative ground)

4. PIKETTY's two "laws of capitalism"

$$r > g?$$

Figure 18 – RATE OF RETURN VS. GROWTH RATE AT THE WORLD LEVEL, ANTIQUITY-2100 (PIKETTY AND SAEZ (2014))



4. PIKETTY's two "laws of capitalism"

PIKETTY and SOLOW

- ▶ Let $g \rightarrow 0$. Then $\sigma_k = s \frac{r}{g} \rightarrow \infty$ for given s and r . That cannot be as $\sigma_k \in [0, 1]$
- ▶ There must be some extra equations that we do not see that link s, r and g .
- ▶ This view that capital income could absorb all income when g goes to zero seems to contradict what we know from balanced growth in the SOLOW model, that is the existence of a Balanced Growth Path.
- ▶ More specifically, going back to the two laws of capitalism :

4. PIKETTY's two "laws of capitalism"

Second law

- The second law states :

$$\frac{k}{y} = \frac{s}{g}$$

- × *Is that reasonable to assume that s stays constant ?*
- × *Are we talking about net output or gross output ?*
- × The distinction is important

$$y^{\text{net}} = y^{\text{gross}} - \delta k$$

- × When k becomes large, δk also becomes large, so that y^{net} becomes low.

4. PIKETTY's two "laws of capitalism"

First law

- First law states :

$$\sigma_k = r \times \frac{k}{y}$$

- × We know that if the production function is Cobb-Douglas $y = Ak^\alpha \ell^{1-\alpha}$ and if there is perfect competition, then $\sigma_k = \alpha$. (see next slide)
- × This means that when $\frac{k}{y}$ increases by 1%, r decreases by 1%.
- × *Is that reasonable to assume that r does not decrease (or does not decrease a lot) when $\frac{k}{y}$ increases?*
- × We will show that this depends on the elasticity of substitution between capital and labor.

4. PIKETTY's two "laws of capitalism"

Cobb-Douglas and perfect competition

- Representative firm maximizes profit :

$$\max_{k,\ell} \Pi = \underbrace{Ak^\alpha \ell^{1-\alpha}}_y - rk - w\ell$$

- First order conditions are :

$$\begin{aligned} /k : \quad \alpha \underbrace{Ak^{\alpha-1} \ell^{1-\alpha}}_{y/k} &= r \\ / \ell : \quad (1-\alpha) \underbrace{Ak^\alpha \ell^{-\alpha}}_{y/\ell} &= w \end{aligned}$$

- and therefore

$$\begin{aligned} \sigma_k &= \frac{rk}{y} = \alpha \\ \sigma_\ell &= \frac{w\ell}{y} = 1 - \alpha \end{aligned}$$

4. PIKETTY's two "laws of capitalism"

Questions

- ▶ We will first discuss on the assumption under which $\frac{k}{y}$ becomes arbitrarily large when g goes to 0.
- ▶ We then discuss the role of the elasticity of substitution between capital and labor.
- ▶ We will do that in the SOLOW model, which is the benchmark neoclassical model of growth.

5. $\frac{k}{y}$ in the SOLOW model and beyond

The SOLOW model

► In levels

$$Y_t = C_t + I_t$$

$$K_{t+1} = (1 - \delta)K_t + I_t$$

$$Y_t = F(K_t, z_t L_t)$$

$$I_t = sY_t$$

$$L_t = (1 + g_L)^t L_0$$

$$z_t = (1 + g_z)^t z_0$$

- F is constant returns to scale, so that $\frac{Y_t}{z_t L_t} = \frac{F(K_t, z_t L_t)}{z_t L_t} = F\left(\frac{K_t}{z_t L_t}, \frac{z_t L_t}{z_t L_t}\right)$
 $= F\left(\frac{K_t}{z_t L_t}, 1\right)$

5. $\frac{k}{y}$ in the SOLOW model and beyond

The SOLOW model

- Denoting $x = X/(zL)$ and $(1 + g) = (1 + g_z)(1 + g_L)$

$$\begin{aligned}y_t &= c_t + i_t \\(1 + g)k_{t+1} &= (1 - \delta)k_t + i_t \\y_t &= F(k_t, 1) = f(k_t) \\i_t &= sy_t\end{aligned}$$

- where we have used $K_{t+1} = (1 - \delta)K_t + I_t \Leftrightarrow \frac{K_{t+1}}{z_{t+1}L_{t+1}} \frac{z_{t+1}L_{t+1}}{z_t L_t} = \frac{(1-\delta)K_t + I_t}{z_t L_t} \Leftrightarrow \frac{K_{t+1}}{z_{t+1}L_{t+1}} \frac{z_{t+1}L_{t+1}}{z_t L_t} = (1 - \delta)\frac{K_t}{z_t L_t} + \frac{I_t}{z_t L_t} \Leftrightarrow (1 + g)k_{t+1} = (1 - \delta)k_t + i_t$.

5. $\frac{k}{y}$ in the SOLOW model and beyond

The SOLOW model

- ▶ The model collapses to : $(1 + g)k_{t+1} = (1 - \delta)k_t + s \underbrace{f(k_t)}_{y_t},$

- ▶ so that at the steady state – i.e. when $k_{t+1} = k_t = k \ \forall t$:

$$\frac{k}{y} = \frac{s}{g + \delta}$$

- ▶ Note that this is not PIKETTY second law :

$$\frac{k}{y} = \frac{s}{g}$$

- ▶ We see that in the SOLOW model, $\frac{k}{y}$ is not infinite when $g = 0$.
- ▶ Is PIKETTY wrong ? (under the assumption that the SOLOW model is the truth)
- ▶ No : As we will see, PIKETTY is considering *net* output y^N , not gross output as in SOLOW
- ▶ What are the implications of the SOLOW model for $\frac{k}{y^N}$?

5. $\frac{k}{y}$ in the SOLOW model and beyond

Net and gross accounting

- ▶ Define net output as $y^N = y - \delta k$
- ▶ PIKETTY assumption is that investment is a constant fraction of net output.
- ▶ In PIKETTY, it is the net saving rate s^N that is constant, whereas it is the gross one s in SOLOW.
- ▶ Does it make a difference?
- ▶ What is the implication of the SOLOW model for s^N ?
- ▶ What is the dynamics of the SOLOW model when s^N is constant instead of s ?
- ▶ Let's solve the SOLOW model steady state when either net or gross saving rate is constant.

5. $\frac{k}{y}$ in the SOLOW model and beyond

The two laws of capitalism again

- ▶ Take first the plain SOLOW model where $s = k/y$ is constant.
- ▶ With our notations, PIKETTY's laws are

$$\sigma_k^N = r \times \frac{k}{y^N}$$

and

$$\frac{k}{y^N} = \frac{s^N}{g}$$

- ▶ Let's find the expression for s^N .

5. $\frac{k}{y}$ in the SOLOW model and beyond

The net saving rate in SOLOW

- ▶ By definition : $s^N = \frac{i^N}{y^N} = \frac{sy - \delta k}{y - \delta k} = \frac{s - \delta k/y}{1 - \delta k/y}$
- ▶ Along a BGP, we have shown that $\frac{k}{y} = \frac{s}{g + \delta}$.
- ▶ Therefore,

$$s^N = \frac{gs}{g + \delta(1 - s)}$$

- ▶ Comments :
 - × s^N is not constant when g varies $\leadsto$ the PIKETTY model is not the SOLOW model.
 - × The only case where the two models are similar is when $s = 1$. In that case, s^N does not depend on g , but this is not a realistic case.
 - × When $g = 0$, $s^N = 0$.
 - × One cannot take the second law of capitalism and reduce g keeping s^N constant if the economy is well described by the SOLOW model.

$$\frac{k}{y^N} = \frac{s^N(g)}{g}$$

- × In the SOLOW model, the net saving rate is not constant but a function of g :
 $s^N = s^N(g)$

5. $\frac{k}{y}$ in the SOLOW model and beyond

When $g \rightarrow 0$

$$s^N = \frac{gs}{g + \delta(1 - s)}$$

$$\frac{k}{y^N} = \frac{s^N(g)}{g}$$

- Therefore, when $g \rightarrow 0$

$$\frac{k}{y^N} = \frac{s}{\delta(1 - s)}$$

- Under the assumptions of the SOLOW model, $\frac{k}{y^N} \rightarrow \frac{s}{\delta(1-s)}$ (and not $+\infty$) when $g \rightarrow 0$.
- Gross saving rate in 2019 : 18.2% (USA), 16.4% (UK), 26.2% (Germany), 21.8% (France), 24.7% (Japan).
- Assume $s = 20\%$ and $\delta = 4\%$. Then, when $g \rightarrow 0$, $\frac{k}{y^N} \rightarrow 6.25$.

5. $\frac{k}{y}$ in the SOLOW model and beyond

SOLOW with constant net saving rate

- ▶ Assume now that the net saving rate is constant. (call it the “PIKETTY-meets-SOLOW” model)
- ▶ This means that $i_t^N = s^N y_t^N$, where $i_t^N = i_t - \delta k_t$.
- ▶ Therefore

$$\begin{aligned}(1 + g)k_{t+1} &= (1 - \delta)k_t + i_t \\ &= k_t + i_t^N = k_t + s^N y_t^N\end{aligned}$$

- ▶ Therefore along a BGP

$$\frac{k}{y^N} = \frac{s^N}{g}$$

- ▶ We have PIKETTY second law (nothing is wrong with it).
- ▶ Therefore k tend to be infinitely large with respect to y^N when g goes to zero.
- ▶ But this is because $y^N \rightarrow 0$
- ▶ What is the value of s in such a case?
- ▶ From the model with s^N constant, we can deduce the level of s . Let's do it.

5. $\frac{k}{y}$ in the SOLOW model and beyond

SOLOW with constant net saving rate

- ▶ We first compute $\frac{y^N}{y}$
- ▶ we have

$$\begin{aligned}(1 + g)k_{t+1} &= k_t + s^N y_t^N \\ &= k_t + s^N (y_t - \delta k_t)\end{aligned}$$

- ▶ Along a BGP, $gk = \underbrace{s^N (y - \delta k)}_{y^N}$, which implies

$$y = \frac{g + s^N \delta}{s^N} k$$

- ▶ Given that $y^N = \frac{g}{s^N} k$, we have

$$\frac{y^N}{y} = \frac{g}{g + s^N \delta}$$

5. $\frac{k}{y}$ in the SOLOW model and beyond

SOLOW with constant net saving rate

- The gross saving rate is

$$s = 1 - \frac{c}{y} = 1 - \underbrace{\frac{c}{y^N}}_{1-s^N} \times \underbrace{\frac{y^N}{y}}_{\frac{g}{g+s^N\delta}}$$

$$s = \frac{s^N(g + \delta)}{g + s^N\delta}$$

- We see that s is indeed a function of g
- When $g = 0$, a constant s^N (what PIKETTY is assuming) implies that $s = 1$, which is highly implausible.

5. $\frac{k}{y}$ in the SOLOW model and beyond

Conclusion

- ▶ In the SOLOW model, if the saving rate is constant, then k/y does not go to infinity when g goes to zero
- ▶ In the PIKETTY-meets-SOLOW model, if the net saving rate is constant, then k/y^N can go to infinity when g goes to zero, but at the same time the saving rate s goes to one, which is implausible
- ▶ PIKETTY's second law cannot be used to study what happens when g goes to zero if the economy is well described by the SOLOW model.

6. The elasticity of substitution between k and ℓ

Back to σ_k

- ▶ Question : when k/y increases, does r decrease a lot or not?
- ▶ $\sigma_k = r \frac{k}{y}$
- ▶ If r decrease a lot, then σ_k will not increase when k/y increase
- ▶ Let σ be the elasticity of substitution between k and ℓ
- ▶ One can show that the elasticity ν of σ_k w.r.t k/y is

$$\nu = 1 - \frac{1}{\sigma}$$

- ▶ Therefore,
 - × if $\sigma = 1$ (Cobb-Douglas), σ_k is constant
 - × if $\sigma < 1$ (k and ℓ are complement), σ_k decreases with k/y
 - × if $\sigma > 1$ (k and ℓ are substitute), σ_k increases with k/y

6. The elasticity of substitution between k and ℓ

Computation Details

Let the production function be CES : $y = F(k, \ell) = \left(\underbrace{ak^\epsilon + b\ell^\epsilon}_X \right)^{\frac{1}{\epsilon}}$. From the profit maximization FOC, we have $F'_k = ak^{\epsilon-1}X^{\frac{1}{\epsilon}-1} = r$ and $F'_\ell = b\ell^{\epsilon-1}X^{\frac{1}{\epsilon}-1} = w$. Taking the ratio of those two FOCs implies $\frac{k}{\ell} = \left(\frac{b}{a} \frac{r}{w} \right)^{\frac{1}{\epsilon-1}}$. From that equation, we can compute the elasticity of substitution between k and ℓ : $\sigma = -\frac{\partial(\frac{k}{\ell})/\left(\frac{k}{\ell}\right)}{\partial(\frac{r}{w})/\left(\frac{r}{w}\right)} = -\frac{\partial(\frac{k}{\ell})}{\partial(\frac{r}{w})} \times \frac{r/w}{k/\ell} = -\frac{1}{\epsilon-1}$. Let $\sigma_k = r \frac{k}{y}$ be the share of capital in total income. Using the FOC for capital and the expression of y , we have $\sigma_k = ak^{\epsilon-1}X^{\frac{1}{\epsilon}-1} \left(\frac{k}{X^{\frac{1}{\epsilon}}} \right) = a \left(\frac{k}{y} \right)^\epsilon$. Therefore, the elasticity ν of σ_k w.r.t k/y is $\nu = \epsilon = 1 - \frac{1}{\sigma}$.

6. The elasticity of substitution between k and ℓ

Back to σ_k

- ▶ To compare to PIKETTY, we need the elasticity of substitution σ^N for net output.
- ▶ One can show (algebra is a bit tedious, let's not prove it) :

$$\sigma^N = \underbrace{\frac{F_k - \delta}{F_k}}_A \times \underbrace{\frac{y}{y - \delta k}}_B \times \sigma$$

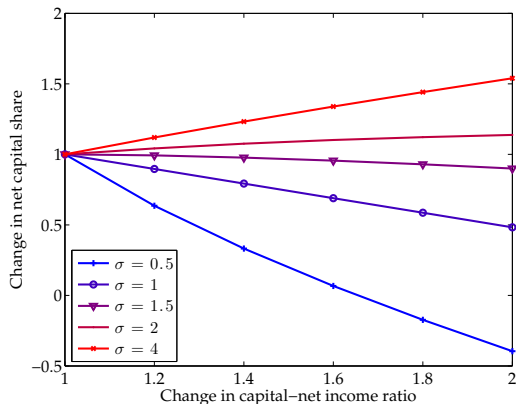
- ▶ $\frac{F_k - \delta}{F_k}$ is the ratio of net to gross return to capital.
 - × $F_k - \delta \approx 5\%$
 - × $\delta \approx 4\%$
 - × Therefore $A \approx .55$
- ▶ $\frac{y}{y - \delta k}$ is the ratio of gross to net output
 - × $\frac{\delta k}{y} \approx 15.8\%$ (USA, 2013)
 - × Therefore $B \approx 1.19$
- ▶ In total, $\sigma^N = .66\sigma$

6. The elasticity of substitution between k and ℓ

Back to σ_k

- Using these formulas, Figure 19 shows how the net capital share of income changes as we increase the ratio k/y^N , for various σ .

Figure 19 – NET CAPITAL-INCOME RATIO AND NET CAPITAL SHARE, BY GROSS ELASTICITY σ . (ROGNLIE (2014))



6. The elasticity of substitution between k and ℓ

Back to σ_k

- ▶ In the data, it is hard to find estimates of σ that are larger than 2.
- ▶ Therefore, for plausible σ , an increase in k/y does not lead to a big increase in the capital income share in the long run (and likely a decline)

7. Summary

- ▶ News set of fact brought by PIKETTY about income distribution
- ▶ We observe an increase in income inequalities and wealth-to-income ratios
- ▶ PIKETTY extrapolates an explosion of k/y and inequalities when $r > g$ and g low.
- ▶ Using the SOLOW model with constant net saving rate or constant saving rate, this is unlikely.
- ▶ Using reasonable estimates of elasticity of substitution between k and ℓ , it is also unlikely.
- ▶ This does not mean that inequalities are at the *right* level (to be defined), not that they cannot stay high for a long period (but not along a BGP) if the SOLOW model is a good description of the data.

8. References

- ▶ Auten Gerald and David Splinter, *"Income Inequality in the United States : Using Tax Data to Measure Long-Term Trends"*, forthcoming in the Journal of Political Economy, 2024
- ▶ Bengtsson, Erik and Daniel Waldenström, *"Capital Shares and Income inequality : Evidence from the Long Run"*, IZA working paper, 2015
- ▶ Krusell, Per and Anthony Smith, *"Is Piketty's "Second Law of Capitalism" Fundamental ?"*, working paper, 2014
- ▶ Piketty, Thomas, *"Capital in the Twenty-First Century"*, Harvard University Press, 2014
- ▶ Piketty, Thomas and Emmanuel Saez, *"Inequality in the long run"*, Science, Vol. 244 Iss. 6186, 2014
- ▶ Rognlie, Matthew, *"A note on Piketty and diminishing returns to capital"*, working paper, 2014
- ▶ Sandra, Agnar, *"The Principal Problem in Political Economy : Income Distribution in the History of Economic Thought"*, working paper, 2013